Endpoint residual bounds and fixed step benchmarks for flexible extra anchored gradient

Abstract

For the flexible extra-anchored-gradient method of Boţ and Chenchene, we consider the endpoint schedule $\varepsilon_k = 1/[\theta(k+\beta)]$. A signed quadratic estimate upgrades the established residual bound $O(k^{-1+\delta})$, valid for every $\delta > 0$, to $O(k^{-1})$. The method and its step-size range remain unchanged. The argument uses the prior weaker estimate only to make an error series summable. A scalar strongly monotone example shows that $o(k^{-1})$ cannot hold in general. We also give a fully explicit, conservative bound in terms of the parameters and initial and anchor distances to a chosen zero. These distances need not be known by the algorithm. For initial anchoring and $\beta = 1$, we also give finite-dimensional fixed-step benchmarks: a horizon-dependent skew family forces a normalized residual constant of at least 2, and a five-step scalar estimate yields a uniform fixed-step lower constant $5/2$ in the L-normalized scale, compared with the known full-step FEG constant 2. No matching sharp upper constant or broader parameter comparison is claimed.

Keywords: monotone operator, anchored extragradient, residual convergence

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1 The endpoint question

Let $M : \mathcal{H} \to \mathcal{H}$ be monotone and L-Lipschitz on a real Hilbert space, and suppose $\operatorname{zer} M \neq \varnothing$. Fix $\theta > 0$, $\tau := \theta L < 1$, $\beta > 0$, an anchor $v \in \mathcal{H}$ and an initial point $x_0 \in \mathcal{H}$. The endpoint form of the flexible extra-anchored-gradient iteration is

$$\begin{aligned} y_{k+1} &= x_k - \frac{x_k - v}{k+\beta} - \theta M(x_k), \\ x_{k+1} &= \frac{x_k + v/(k+\beta+1) - \theta M(y_{k+1})}{1 + 1/(k+\beta+1)}, \qquad k \geq 0. \end{aligned} \tag{1}$$

This is Algorithm 1.1 of [1] with $\varepsilon_k = 1/[\theta(k+\beta)]$. Its predictor and corrector have different anchor weights, so we retain the exact coefficients throughout.

Theorem 2.4 and Corollary 2.6 of [1] establish boundedness, strong convergence to the projection of v onto $\operatorname{zer} M$, and

$$\|M(x_k)\| = O(k^{-1+\delta}) \qquad \text{for every } \delta \in (0,1). \tag{2}$$

For larger anchoring coefficient $\alpha > 1$, their corollary already gives $O(k^{-1})$. Their Remark 1.5 explains how the critical continuous-time rate can be recovered while noting the difficulty of transferring the corresponding calculation to discrete time. We use the summability consequence of (2) to close this discrete endpoint.

Theorem 1. *Under the assumptions above, the unchanged iteration* (1) *satisfies*

$$\|M(x_k)\| = O(k^{-1}).$$

The constant may depend on the problem, parameters and initialization. In general the conclusion cannot be strengthened to $o(k^{-1})$.

2 A signed quadratic estimate

Write $t = k + \beta$, $X_k = x_k - v$, and at a fixed step set

$$p = \theta M(x_k), \qquad q = \theta M(y_{k+1}), \qquad r = \theta M(x_{k+1}).$$

The translated update is exactly

$$y_{k+1} - v = (1 - 1/t)X_k - p, \qquad X_{k+1} = \frac{t+1}{t+2}(X_k - q). \tag{3}$$

Define the signed local energy

$$E_k = (t+1)^2 \|p\|^2 + 2(t+1)\langle p, X_k\rangle. \tag{4}$$

Quadratic residual and anchor cross terms are classical in extra-anchored-gradient analysis [3]. The point here is the estimate for the exact coefficients in (1) and the summability bootstrap below, rather than a new general energy principle.

Lemma 1. *For every* $k \geq 0$,

$$E_{k+1} - E_k \leq \|(t+2)r - (t+1)q\|^2 - (t+1)^2 \|p - q\|^2. \tag{5}$$

Moreover,

$$\|(t+2)r - (t+1)q\| \leq \tau(t+2)\|p - q\| + (1+\tau)\|q\| + \frac{2\tau}{t}\|X_k\|. \tag{6}$$

Proof. The exact difference between the right-hand side of (5) and $E_{k+1} - E_k$ is

$$2(t+1)(t+2)\langle r - p, x_{k+1} - x_k\rangle,$$

which is nonnegative by monotonicity. This identity follows by expanding (4) and using $(t+2)X_{k+1} = (t+1)(X_k - q)$.

For the second estimate, (3) gives

$$(t+2)(x_{k+1} - y_{k+1}) = (t+2)(p - q) + q + \frac{2}{t}X_k.$$

Since θM is τ-Lipschitz, the triangle inequality applied to $(t+2)(r-q)+q$ proves (6). ∎

Set

$$C_q = \frac{2(1+\tau^2)(1+\tau)^2}{1-\tau^2}, \qquad C_x = \frac{8\tau^2(1+\tau^2)}{1-\tau^2}. \tag{7}$$

For $0 < \tau < 1$, Young's inequality applied to (6) gives

$$\|(t+2)r - (t+1)q\|^2 \le \frac{1+\tau^2}{2}(t+2)^2 \|p-q\|^2 + C_q \|q\|^2 + \frac{C_x}{t^2} \|X_k\|^2 .$$

Indeed, use the coefficient $\nu = (1+\tau^2)/(2\tau^2) > 1$ on the first squared term, then bound the remaining square by twice the sum of squares. If $t \ge 4/(1-\tau^2)$, then

$$\frac{1+\tau^2}{2}(t+2)^2 \le (t+1)^2.$$

Therefore

$$E_{k+1} \le E_k + C_q \|\theta M(y_{k+1})\|^2 + \frac{C_x}{(k+\beta)^2} \|x_k - v\|^2 \tag{8}$$

on this tail. This is an estimate with summable errors, not an assertion that E_k is nonnegative or nonincreasing.

3 Summability and the endpoint rate

Proof of Theorem 1. If $L = 0$, M is constant and the existence of a zero forces $M = 0$, so the conclusion is immediate. Assume $\tau > 0$.

Boundedness and (2) with $\delta = 1/4$ give

$$\sup_k \|X_k\| < \infty, \qquad \sum_{k=0}^{\infty} \|\theta M(x_k)\|^2 < \infty.$$

Furthermore, $y_{k+1} - x_k = -p - X_k/t$, and Lipschitz continuity gives

$$\|q\| \le (1+\tau)\|p\| + \frac{\tau}{t}\|X_k\| .$$

Consequently $\sum_k \|q\|^2 < \infty$ as well. Both error series in (8) are finite, so E_k is bounded above. This use of the prior result is noncircular: only the already established exponent $-3/4$ and boundedness are used.

Fix any $x^* \in \operatorname{zer} M$ and let $X^* = x^* - v$. Monotonicity implies $\langle p, X_k - X^* \rangle \ge 0$, and hence

$$E_k \ge \|(t+1)p + X^*\|^2 - \|X^*\|^2 . \tag{9}$$

The upper bound on E_k therefore bounds $(t+1)\|p\|$, proving the asserted $O(k^{-1})$ rate. ∎

For clarity about constants, choose K so that $K + \beta \ge 4/(1-\tau^2)$ and define the finite quantity

$$B = E_K + C_q \sum_{j=K}^{\infty} \|\theta M(y_{j+1})\|^2 + C_x \sum_{j=K}^{\infty} \frac{\|x_j - v\|^2}{(j+\beta)^2}.$$

Then (9) yields, for $k \geq K$,

$$\|M(x_k)\| \leq \frac{\sqrt{B + \|X^*\|^2} + \|X^*\|}{\theta(k+\beta+1)}. \tag{10}$$

Here B is expressed through prior orbit bounds and convergent series. Section 5 gives a separate explicit estimate using theoretical initial and anchor radii. Formula (10) does not identify an optimal initial-data-only worst-case constant. Section 6 supplies a separate fixed-step lower comparison with fast extragradient; it does not determine the optimal upper constant or settle the broader parameter question in [1].

4 Order sharpness

Take $\mathcal{H} = \mathbb{R}$, $M(x) = x/2$, $\theta = 1$, $\beta = 1$, $v = 1$ and $x_0 = 1$. The unique zero is 0 and $\theta L = 1/2$. With $t = k + 1$, direct substitution in (1) gives

$$x_{k+1} = \frac{t+1}{t+2}\left[\left(\frac{3}{4} + \frac{1}{2t}\right)x_k + \frac{1}{t+1} - \frac{1}{2t}\right].$$

Set $s_k = (k+1)M(x_k) = tx_k/2$. Then

$$s_{k+1} = a_t s_k + b_t, \quad a_t = \frac{(t+1)^2(3t+2)}{4t^2(t+2)}, \quad b_t = \frac{t^2-1}{4t(t+2)}.$$

Since $a_t \to 3/4$ and $b_t \to 1/4$, a scalar contraction comparison gives $s_k \to 1$. Equivalently,

$$(k+1)M(x_k) \longrightarrow 1.$$

Thus the residual is not $o(k^{-1})$, even for a scalar strongly monotone linear operator. This establishes order sharpness, without asserting an optimal uniform constant.

5 An explicit bound in terms of theoretical radii

The orbit sums in (10) can be bounded without using a prior asymptotic convergence result. The formulas below are deliberately conservative. Their inputs include distances to a chosen zero, which need not be known by the algorithm; this is a theoretical uniform estimate, not a solution-free stopping certificate.

Assume $0 < \tau < 1$, fix $x^* \in \operatorname{zer} M$, and put

$$D = \|x_0 - x^*\|, \qquad d = \|x^* - v\|, \qquad \widehat{F}(X) = \theta M(v + X), \qquad z = x^* - v.$$

Thus $\widehat{F}$ is monotone and τ-Lipschitz, $\widehat{F}(z) = 0$, and $X_k = x_k - v$ obeys (3). Define, for $t > 0$,

$$a(t) = \frac{t+1}{t+2}, \quad A(t) = a(t)\left(1 + \frac{\tau}{t}\right), \quad B(t) = 1 - a(t) + a(t)\frac{\tau}{t}.$$

Set

$$T_0 = \max\left\{1, \frac{2\tau}{1-\tau}\right\}, \qquad K_0 = \max\{0, \lceil T_0 - \beta \rceil\}.$$

Starting from $\bar{e}_0 = D$, compute only the finite prefix

$$\bar{e}_{j+1} = A(j+\beta)\bar{e}_j + B(j+\beta)d, \qquad 0 \le j < K_0.$$

Define the two radii

$$\mathcal{E} = \max\left\{\max_{0\le j\le K_0} \bar{e}_j, \frac{3+\tau}{1-\tau}d\right\}, \qquad R = \mathcal{E} + d. \tag{11}$$

Next put

$$\begin{aligned} C_0 &= \frac{1+\tau^4}{1-\tau^2}, & \Gamma &= 2 + 4C_0, \\ T &= \max\left\{T_0, 2, 8C_0(2+\tau)^2, \frac{4}{1-\tau^2}\right\}, & K &= \max\{0, \lceil T - \beta \rceil\}, \qquad u = K + \beta. \end{aligned} \tag{12}$$

With $\nu = 5/4$, define

$$\begin{aligned} P_0 &= \frac{1}{2}\left(\tau\mathcal{E} + \frac{R}{u}\right)^2, & A_0 &= \max\{P_0 u^\nu, 4\Gamma R^2 u^{\nu-2}\}, \\ S_\nu &= u^{-\nu} + \frac{u^{1-\nu}}{\nu - 1}, & S_2 &= u^{-2} + u^{-1}, \\ \mathcal{Q} &= 8(1+\tau)^2 A_0 S_\nu + [4(1+\tau)^2 + 2\tau^2]R^2 S_2. \end{aligned} \tag{13}$$

Finally use the constants in (7) to set

$$\begin{aligned} \mathcal{B} &= (u+1)^2\tau^2\mathcal{E}^2 + 2(u+1)\tau\mathcal{E}R + C_q\mathcal{Q} + C_x R^2 S_2, \\ \Lambda &= \max\{(u+1)\tau\mathcal{E}, \sqrt{\mathcal{B} + d^2} + d\}. \end{aligned} \tag{14}$$

All these numbers depend only on τ, β, D, d.

Theorem 2. *The exact iteration* (1) *satisfies, for every* $k \ge 0$,

$$\|M(x_k)\| \le \frac{\Lambda}{\theta(k+\beta+1)}. \tag{15}$$

No prior asymptotic residual estimate is needed for this bound. If $v = x_0$, *then* $D = d$ *and homogeneity reduces its numerator to a function of* τ, β *times that common distance. No optimality of this constant is asserted.*

Proof. We first establish the uniform radius. For the ordinary extragradient map

$$G(X) = X - \widehat{F}(X - \widehat{F}(X)),$$

expansion, monotonicity at z, and Lipschitz continuity give

$$\|G(X) - z\|^2 \le \|X - z\|^2 - (1-\tau^2)\left\|\widehat{F}(X)\right\|^2. \tag{16}$$

Indeed, insert $X - \widehat{F}(X)$ into the inner product in the norm expansion. The remaining terms equal $\left\|\widehat{F}(X) - \widehat{F}(X - \widehat{F}(X))\right\|^2 - \left\|\widehat{F}(X)\right\|^2$.

The actual predictor differs from $X - \widehat{F}(X)$ by $-X/t$. Therefore, with $e_k = \|X_k - z\|$,

$$e_{k+1} \le a(t)e_k + (1 - a(t))d + a(t)\frac{\tau}{t}(e_k + d) = A(t)e_k + B(t)d.$$

For $t \ge T_0$ we have $0 \le A(t) < 1$ and

$$\frac{B(t)}{1 - A(t)} = \frac{(1+\tau)t + \tau}{(1-\tau)t - \tau} \le \frac{3+\tau}{1-\tau}.$$

The finite prefix and induction prove $e_k \le \mathcal{E}$ and $\|X_k\| \le R$ for every k.

Now let $h_k = 1/(k+\beta)$ and introduce

$$r_k = \widehat{F}(X_k) + h_k X_k, \qquad \Psi_k = \tfrac{1}{2}\|r_k\|^2, \qquad W_k = h_{k+1}X_{k+1} - h_k X_k - h_k r_k.$$

We claim the quantitative estimate

$$\Psi_{k+1} - \Psi_k \le -2h_k\Psi_k + (h_{k+1} - h_k)\langle X_{k+1}, r_k\rangle + \frac{C_0}{2}\|W_k\|^2. \tag{17}$$

Here is a direct verification. Suppress subscripts, write $\Delta X = X_{k+1} - X_k$, $\Delta F = \widehat{F}(X_{k+1}) - \widehat{F}(X_k)$, and set

$$N = \widehat{F}(X_k) - h_k r_k - \widehat{F}(y_{k+1} - v).$$

The recurrence gives $\Delta X + r_k = N - W_k$ and $X_{k+1} - (y_{k+1} - v) = N - W_k$. Expanding the residual norm yields

$$\begin{aligned}\Delta\Psi = &-\langle \Delta F, \Delta X\rangle - 2h_k\Psi_k + \Delta h_k\langle X_{k+1}, r_k\rangle \\ &+ \tfrac{1}{2}\left\|\widehat{F}(X_{k+1}) - \widehat{F}(y_{k+1} - v)\right\|^2 - \tfrac{1}{2}\|N\|^2 + \tfrac{1}{2}\|W_k\|^2.\end{aligned}$$

The first inner product is nonpositive after its minus sign. Bound the last line by Lipschitz continuity and Young's inequality with $\delta_1 = (1-\tau^2)/(2\tau^2) > 0$. The coefficient of $\|N\|^2$ is nonpositive, while the coefficient of $\|W_k\|^2$ is

$$\tfrac{1}{2}(1 + \tau^2 + \tau^2/\delta_1) = \tfrac{1}{2}C_0.$$

This proves (17).

The further exact identity

$$(1 + h_k)\Delta X = -\Delta h_k X_{k+1} - [\widehat{F}(y_{k+1} - v) - \widehat{F}(X_k)] - r_k$$

gives

$$\|W_k\| \le 2|\Delta h_k|R + h_k(2+\tau)\|r_k\|.$$

Also,

$$\Delta h_k \langle X_{k+1}, r_k \rangle \le \frac{h_k}{4}\Psi_k + \frac{2(\Delta h_k)^2}{h_k}R^2.$$

Use $|\Delta h_k| \le h_k^2$, $h_k \le 1$, and the threshold in (12). The contribution of $\|W_k\|^2$ to the Ψ_k coefficient is at most $h_k/4$. Consequently, for $k \ge K$,

$$\Psi_{k+1} \le \left(1 - \frac{3}{2t_k}\right)\Psi_k + \frac{\Gamma R^2}{t_k^3}, \qquad t_k = k + \beta. \tag{18}$$

At K, the radius bound gives $\Psi_K \le P_0$. Convexity yields $(1+1/t)^{-\nu} \ge 1 - \nu/t$. Since $3/2 - \nu = 1/4$, the definition of A_0 and induction in (18) imply

$$\Psi_k \le \frac{A_0}{t_k^\nu}, \qquad k \ge K.$$

The coefficients $1 - 3/(2t_k)$ are nonnegative because $t_k \ge 2$. Integral comparison gives

$$\sum_{k=K}^{\infty} \Psi_k \le A_0 S_\nu, \qquad \sum_{k=K}^{\infty} t_k^{-2} \le S_2.$$

Moreover,

$$\left\|\widehat{F}(X_k)\right\|^2 \le 4\Psi_k + 2R^2/t_k^2,$$
$$\left\|\widehat{F}(y_{k+1} - v)\right\|^2 \le 2(1+\tau)^2 \left\|\widehat{F}(X_k)\right\|^2 + 2\tau^2 R^2/t_k^2.$$

Summing these inequalities proves that the second squared residual has sum at most $\mathcal{Q}$ on the tail.

Apply (8). At K, $\|p\| \le \tau\mathcal{E}$, so the signed energy is bounded by the first two terms of $\mathcal{B}$. The two tail sums give $E_k \le \mathcal{B}$ for $k \ge K$. The lower bound (9) then gives

$$\|\theta M(x_k)\| \le \frac{\sqrt{\mathcal{B} + d^2} + d}{t_k + 1}.$$

For $k < K$, use $\|\theta M(x_k)\| \le \tau\mathcal{E}$ and $t_k + 1 \le u + 1$. The maximum defining Λ covers both ranges and proves (15). ∎

The constant may be large, especially as $\tau \uparrow 1$ or for an aggressive initial anchoring parameter. Its purpose is to remove dependence on unquantified orbit bounds while preserving the distinction between an explicit estimate and an optimal complexity constant.

6 Fixed step benchmarks at the endpoint

We now restrict to $\beta = 1$ and $v = x_0$. The comparison in this section is on finite-dimensional real spaces. It is separate from the Hilbert-space upper bounds proved above. The normalized step $\tau = \theta L$ is chosen once for the whole operator class, independently of the instance and the observation horizon.

Translate v to zero and write $F(X) = \theta M(v+X)$, $z = x^* - v$ and $R = \|z\| > 0$. Then F is monotone and τ-Lipschitz, $F(z) = 0$, and the exact iteration is

$$Y_{k+1} = \frac{k}{k+1}X_k - F(X_k), \qquad X_{k+1} = \frac{k+2}{k+3}\big(X_k - F(Y_{k+1})\big), \qquad X_0 = 0. \tag{19}$$

For fixed $\tau \in (0,1)$, define the possibly infinite constants

$$\mathfrak{C}(\tau) = \sup \frac{k\,\|F(X_k)\|}{\|z\|}, \qquad \mathfrak{K}(\tau) = \frac{\mathfrak{C}(\tau)}{\tau}. \tag{20}$$

The supremum is over all finite dimensions, all such operators and chosen nonzero zeros z, and all integers $k \geq 1$. Thus $\mathfrak{K}(\tau)$ is the worst-case constant in the original normalization $k\,\|M(x_k)\|/(LR)$.

Theorem 3. *The exact method* (19) *satisfies the lower bounds*

$$\mathfrak{C}(\tau) \geq 2 \quad (0 < \tau < 1), \qquad \inf_{0<\tau<1} \mathfrak{K}(\tau) \geq \frac{5}{2}. \tag{21}$$

The first assertion is witnessed by a horizon-dependent family of affine skew operators. The second combines that family with a fixed five-step scalar calculation. Neither inequality is an upper bound or a claim that these lower constants are optimal.

For comparison, Lee and Kim's FEG method [2, Algorithm 1 and Theorem 4.1, $\rho = 0$] uses the same initial anchor and, at the full step $1/L$, guarantees

$$k\,\big\|M(x_k^{\mathrm{FEG}})\big\| \leq 2LR \qquad (k \geq 1) \tag{22}$$

on this finite-dimensional monotone class. Their Example 2 attains the bound on a planar skew map. Both methods have a standard budget of at most two new operator evaluations per completed iteration; FEG's first zero-coefficient predictor query may be omitted. Thus FEG receives no additional calls at the same iteration count. Final residual diagnostics, if evaluated, are counted consistently. The comparison is not an oracle-optimality statement.

6.1 A horizon dependent skew family

Identify $\mathbb{R}^2$ with $\mathbb{C}$, set $z = 1$ and consider $F(X) = \lambda(X-1)$. For real $\lambda \geq 0$ this is a scalar monotone map; for $\lambda = i\omega$ it is a planar skew map. Put

$$b = 1 - \lambda + \lambda^2, \qquad p_k = F(X_k), \qquad u_k = (k+2)p_k + 1.$$

Direct substitution in the original two stages gives

$$u_0 = 1 - 2\lambda, \qquad u_{k+1} = \left(b + \frac{\lambda}{k+1}\right) u_k + \frac{b-1}{k+1}. \tag{23}$$

Fix $c > 0$ and a horizon N, and take $\omega = c/N$, $\lambda = i\omega$. For $N \geq c/\tau$ this is an admissible τ-Lipschitz instance, held fixed throughout its own run. We claim

$$NF_N(X_N) \longrightarrow e^{-ic} - 1. \tag{24}$$

Here the operator depends on N; this is not a limit along one fixed operator's infinite trajectory.

To prove the claim uniformly on each finite run, write $H_N = \sum_{j=1}^{N} 1/j$ and $\delta = |b-1| = \omega\sqrt{1+\omega^2}$. For $\omega \leq 1$, $|b|^2 = 1 - \omega^2 + \omega^4 \leq 1$. Iterating (23) and bounding each product by $\exp(\omega H_N)$ gives

$$|u_j| \leq M_N := e^{\omega H_N}\big(\sqrt{1+4\omega^2} + \delta H_N\big) \qquad (0 \leq j \leq N).$$

Comparison with b^j then gives

$$|u_N - b^N| \leq 2\omega + (\omega M_N + \delta)H_N.$$

The Taylor remainder $|e^{-i\omega} - (1 - i\omega)| \leq \omega^2/2$ and a telescoping product yield

$$|b^N - e^{-ic}| \leq \frac{3}{2}N\omega^2.$$

Since $Np_N = N(u_N - 1)/(N+2)$, we obtain the explicit bound

$$|Np_N - (e^{-ic} - 1)| \leq 2\omega + (\omega M_N + \delta)H_N + \frac{3}{2}N\omega^2 + \frac{4}{N+2} = O_c\left(\frac{\log N}{N}\right).$$

This proves (24). Taking $c = \pi$ proves $\mathfrak{C}(\tau) \geq 2$ for every fixed $\tau > 0$. If the Lipschitz constant is required to equal τ exactly, append an unused skew block of modulus τ in $\mathbb{R}^4$ and take $z = (1, 0, 0, 0)$. That block remains zero, so the same witness applies.

The index in this statement matters. For the single fixed map $F(X) = \frac{i}{2}(X - 1)$, exact evaluation of (23) gives

$$6F(X_4) = -\frac{979 + 389i}{512}, \qquad \|6F(X_4)\|^2 = \frac{554881}{131072} > 4.$$

Thus a hypothetical bound with constant 2 cannot replace k by $k + 2$, even at $\tau = 1/2$.

6.2 A uniform fixed step gap

For $a \in [4/5, 1)$, use the scalar map $F_a(X) = a(X - 1)$ and $X_0 = 0$. This map is globally monotone, has exact Lipschitz modulus a and zero $z = 1$. Five exact updates give

$$X_5 = \frac{a(1-a)}{420}P_8(a), \tag{25}$$

where

$$P_8(a) = 120a^8 - 326a^7 + 989a^6 - 1523a^5 + 2476a^4 \\ - 2366a^3 + 2567a^2 - 1337a + 1200.$$

We verify the whole interval, rather than only its endpoints. With $a = (4+s)/5$, direct expansion gives

$$\begin{aligned}820312500\left(\tfrac{1}{2} - X_5\right) = {} & 7647850 + 193956856s + 140162901s^2 \\ & + 49372176s^3 + 15111761s^4 + 3241726s^5 \\ & + 579619s^6 + 75024s^7 + 7799s^8 + 514s^9 + 24s^{10}.\end{aligned}$$

Every coefficient is positive, so $X_5 < 1/2$ for $s \in [0,1]$. Therefore

$$\frac{5\,\|F_a(X_5)\|}{a} = 5(1 - X_5) > \frac{5}{2} \qquad (4/5 \le a < 1).$$

For $0 < \tau < 4/5$, the skew family instead gives $\mathfrak{K}(\tau) \ge 2/\tau > 5/2$. Combining these two ranges proves the second assertion of Theorem 3. We claim only the non-strict infimum inequality in (21), not strictness of that infimum.

The comparison with (22) concerns only the endpoint coefficient, $\beta = 1$, the initial anchor, and one fixed normalized step selected uniformly over the operator class. It does not cover other anchoring parameters, adaptive or horizon-dependent step choices, or per-instance tuning. The factor $5/2$ is a conservative lower benchmark, not a sharp determination of $\mathfrak{K}$. In particular the separate same-step question

$$k\,\|\theta M(x_k)\| \le 2R \quad \text{when } \theta L = 1/2,\ \beta = 1,\ v = x_0$$

is not proved or refuted here. The lower family shows what a matching upper estimate would have to attain.

References

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